

Literature Status: One-Sided James Theorem

FA/Banach run `fa.banach.001`

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Packet Status

This is a `literature_already_answered` packet, not an original partial result from this run. It records that the arbitrary-Banach-space version of the one-sided James compactness theorem asked for by Cascales–Orihuela–Perez is already answered in later literature.

Original Source

The source paper is B. Cascales, J. Orihuela, and A. Perez, *One-sided James’ compactness theorem*, arXiv:1508.00496, published in *Journal of Mathematical Analysis and Applications* 445(2), 1267–1283, 2017, DOI 10.1016/j.jmaa.2016.03.080.

The source proves the relevant two-set theorem under the additional hypothesis that (B_{E^*}, w^*) is convex block compact. Its final remarks ask whether Theorems 2 and 10 remain valid for arbitrary Banach spaces.

Supporting Literature

Orihuela’s later paper *Conic James’ compactness theorem*, arXiv:1610.02763, settles the Theorem 10 direction for arbitrary Banach spaces and records that Theorem 2 remained open at that point.

The decisive later answer is Warren B. Moors, *On a one-sided James’ theorem*, *Journal of Mathematical Analysis and Applications* 449(1), 528–530, 2017, DOI 10.1016/j.jmaa.2016.12.019. Crossref records this paper with that title, author, journal, volume, issue, and page range, and its reference list cites the Cascales–Orihuela–Perez JMAA paper directly.

Literature Status Theorem. *The arbitrary-Banach-space version of the one-sided James compactness theorem from Cascales–Orihuela–Perez is already answered in later literature.*

Status verification. The original source asks whether the theorem can be proved without the convex-block-compactness assumption on the dual ball. Orihuela’s conic James paper handles the companion Theorem 10 direction and leaves Theorem 2 as the remaining issue. Moors’s 2017 note is a separate later paper titled *On a one-sided James’ theorem*; its bibliographic metadata identifies the exact target paper as a cited reference. The later Moors paper therefore turns the target from a pushable partial problem into an already-known literature answer. $\square$

Preserved Partial Work

The previous partial packet is preserved in `partial_reduction_history/`. It contains useful correct reductions and positive families, including the one-set reduction, the affine-slice subcase, conic-dominated families using Orihuela’s theorem, and frustum or finite-dimensional cylinder examples. These should be treated as explanatory material, not as a current open partial to push.

Access Limitation

The anonymous ScienceDirect request available during this update returned HTML or metadata rather than a usable local PDF for Moors’s paper. The packet therefore stores Crossref metadata for the decisive DOI instead of a local Moors PDF. A human with institutional access can add the PDF later.

Human Review Recommendation

Classify this target as `literature_known` / `literature_already_answered`. Do not spend another model round trying to extend the partial result unless the goal is a self-contained expository reproof.